

Interference

Q1. White light falls normally on a film of soapy water of thickness 5×10^{-5} cm and refractive index 1.33. Which wavelength in the visible region will be reflected most strongly?

Given: -

$$\mu = 1.33, t = 5 \times 10^{-5} \text{ cm}, i = 0 \therefore r = 0;$$

Formula: - condition for Constructive interference

$$2\mu t \cos r = (2n-1)\lambda/2$$

Solution: -

$$2\mu t \cos r = (2n-1)\lambda/2$$

$$\begin{aligned}\lambda &= \frac{4\mu t}{(2n-1)} \\ &= \frac{4 \times 1.33 \times 5 \times 10^{-7}}{(2n-1)} \\ &= \frac{2.66 \times 10^{-6}}{(2n-1)}\end{aligned}$$

$$\text{For } n=1 \quad \lambda_1 = 26,660 \text{ \AA}$$

$$\text{For } n=2 \quad \lambda_2 = 8,866 \text{ \AA}$$

$$\text{For } n=3 \quad \lambda_3 = 5,320 \text{ \AA}$$

$$\text{For } n=4 \quad \lambda_4 = 3,800 \text{ \AA}$$

Here $\lambda_3 = 5,320 \text{ \AA}$ lies in the visible region ranging

From $4000 \text{ \AA}$ to $7000 \text{ \AA}$.

Ans: hence, most strongly reflected wavelength is $5,320 \text{ \AA}$

Q2. Fringes of equal thickness are observed in a thin glass wedge of refractive index 1.52. The fringe spacing is 1mm and the wavelength of light is $5893 \text{ \AA}$. Calculate the angle of wedge in second of an arc.

Given: - $\mu=1.52$, $\lambda = 5893 \times 10^{-8} \text{ cm}$,

$$\beta = 1 \text{ mm} = 0.1 \text{ cm}$$

Formula: - the fringe spacing is

$$\beta = \frac{\lambda}{2\mu\theta}$$

Solution: - The angle of wedge in radians is

$$\theta = \frac{\lambda}{2\beta\mu} = \frac{5893 \times 10^{-8} \times 180}{2 \times 1.52 \times 0.1 \times \pi} \times 3600 \text{ second}$$

$$\theta = 40 \text{ seconds of an arc.}$$

Ans: - The angle of wedge is 40 seconds.

Q3. A beam of parallel rays is incident on a film of thickness $4 \times 10^{-5} \text{ cm}$ and of refractive index 1.5 at an angle of incidence of 40° . Show that the wavelength $7.228 \times 10^{-5} \text{ cm}$ will be strongly reflected.

Given: -

$$\mu=1.5, t=4 \times 10^{-5} \text{ cm}, i=40$$

Formula: - condition for Constructive interference

$$2\mu t \cos r = (2n-1)\lambda/2$$

condition for Destructive interference

$$2\mu t \cos r = n\lambda$$

Solution: - $\sin r = \frac{\sin(i)}{\mu} = \frac{\sin 40}{1.5} = 0.4285$

$$\therefore \cos r = 0.9035$$

Let us assume that

$$2\mu t \cos r = m \frac{\lambda}{2}$$

Where if $m = \text{odd}$, its constructive

if $m = \text{even}$, its destructive

$$\begin{aligned} m &= \frac{4\mu t \cos r}{\lambda} \\ &= \frac{4 \times 1.5 \times 4 \times 10^{-5} \times 0.9035}{7.228 \times 10^{-5}} \\ m &= 3 \end{aligned}$$

Ans: As m is odd, the film appears bright in the reflected light

Q4. A soap film of refractive index 1.33 and thickness 1.5×10^{-4} cm is illuminated by white light incident at an angle of 45° . In the reflected light a dark band is observed for the wavelength 5×10^{-5} cm. Calculate the order of the interference band.

Given: -

$$\mu = 1.33, t = 1.5 \times 10^{-4} \text{ cm}, i = 45^\circ$$

Formula: -

condition for Destructive interference

$$2\mu t \cos r = n\lambda$$

Solution: - $\sin r = \frac{\sin(i)}{\mu} = \frac{\sin 45}{1.33} = 0.5317$

$$\therefore \cos r = 0.8469$$

$$\begin{aligned} n &= \frac{2\mu t \cos r}{\lambda} \\ &= \frac{2 \times 1.33 \times 1.5 \times 10^{-4} \times 0.8469}{5 \times 10^{-5}} \end{aligned}$$

$$n = 6.758$$

this shows the highest possible order of the visible band is 6

Ans: hence order of interference band is 6

Q5. A soap film of refractive index 1.33 is illuminated by the normal incident of a beam of monochromatic light of wavelength $5893 \text{ \AA}$.

What is the minimum thickness of the film that will appear (i) dark and (ii) bright in the reflected light?

Given: -

$$\mu=1.33, \lambda=5893 \text{ \AA}, i=0 \therefore r=0$$

Formula: - condition for Constructive interference or brightness

$$2\mu t \cos r = (2n-1)\lambda/2$$

condition for Destructive interference or darkness

$$2\mu t \cos r = n\lambda$$

Solution: -

For dark film,

$$t = \frac{n\lambda}{2\mu}$$

$$t_{\min} = \frac{\lambda}{2\mu} \quad \text{as } n=1$$

$$t_{\min} = \frac{5893 \times 10^{-10}}{2 \times 1.33} = 2215 \text{ \AA}$$

for bright film,

$$t = (2n+1) \frac{\lambda}{4\mu}$$

$$t_{\min} = \frac{\lambda}{4\mu} \quad \text{as } n=1$$

$$t_{\min} = \frac{5893 \times 10^{-10}}{4 \times 1.33} = 1107 \text{ \AA}$$

Ans: minimum thickness for dark film is 2215 \AA

minimum thickness for bright film is 1107 \AA

Q6. A plane wave of monochromatic light falls normally on a uniform thin film of oil which covers a glass plate. The wavelength of the source can be varied continuously. Complete destructive interference is obtained only for wavelengths 5000 \AA and 7000 \AA and for no other wavelengths in between. Find the thickness of the oil layer. Given, refractive index of oil 1.3 and refractive index of glass = 1.5.

Given: -

$$\mu_{\text{oil}} = 1.3, \mu_{\text{glass}} = 1.5, \mu_{\text{air}} = 1, \\ \lambda_1 = 5000 \text{ \AA}$$

$$\lambda_2 = 7000 \text{Å}$$

$$i=0 \therefore r=0$$

Formula: - for Destructive interference

$$\text{Path difference} = (2n-1)\lambda/2$$

As the additional P.D of $\lambda/2$ is introduced at each of the points A and B on the denser surface, the condition becomes:

$$2\mu_{\text{oil}} t_{\text{oil}} \cos r \pm \lambda/2 + \lambda/2 = (2n \pm 1)\lambda/2$$

$$\therefore 2\mu_{\text{oil}} t_{\text{oil}} \cos r = (2n \pm 1)\lambda/2$$

Solution: -

In the formula

$$2\mu_{\text{oil}} t_{\text{oil}} \cos r = (2n \pm 1)\lambda/2$$

$$\lambda = \frac{2}{(2n \pm 1)} 2\mu_{\text{oil}} t_{\text{oil}}$$

$$\text{For } \lambda_1 = 5000 \text{Å}, \quad n = m+1$$

$$\lambda_2 = 7000 \text{Å}, \quad n = m$$

$$\lambda_2 = \frac{2}{(2m \pm 1)} 2\mu_{\text{oil}} t_{\text{oil}}$$

$$\lambda_1 = \frac{2}{(2m \pm 3)} 2\mu_{\text{oil}} t_{\text{oil}}$$

$$\frac{\lambda_2}{\lambda_1} = \frac{(2m \pm 3)}{(2m \pm 1)}$$

$$(2m \pm 1) \lambda_2 = (2m \pm 3) \lambda_1$$

$$2m(\lambda_2 - \lambda_1) = (3\lambda_1 - \lambda_2)$$

$$m = \frac{(3\lambda_1 - \lambda_2)}{2(\lambda_2 - \lambda_1)}$$

$$m = \frac{[(3 \times 5000) - 7000] \times 10^{-8}}{2(7000 - 5000) \times 10^{-8}}$$

$$m = 2$$

using this, it is found that

$$t_{\text{oil}} = \frac{(2m+1)\lambda_2}{4\mu_{\text{oil}}}$$

$$= \frac{5 \times 7000 \times 10^{-10}}{4 \times 1.3} = 6730 \text{ Å}$$

Ans: thickness of oil layer = 6730 Å

Q7. White light is incident on a soap film ($\mu = 1.33$) at an angle 53° and the reflected light shows two consecutive dark bands corresponding to the wavelengths 6100 Å and 6000 Å. Calculate the thickness of the film.

Given: -

$$\mu_{\text{film}} = 1.33, i = 53^\circ$$

$$\lambda_1 = 6100 \text{ Å}$$

$$\lambda_2 = 6000 \text{ Å}$$

Formula: - in this case,

condition for Destructive interference or darkness

$$2\mu t \cos r = n\lambda$$

Solution: - $\sin r = \frac{\sin(i)}{\mu} = \frac{\sin 53}{1.33} = 0.6$

$$\therefore r = 36.85^\circ$$

From the formula, it is found that the values of μ, r and t are given

Hence

$$n\lambda = 2\mu t \cos r = \text{constant}$$

$$n\lambda = \text{constant}$$

$$n \propto \frac{1}{\lambda}$$

$\therefore$ if the consecutive bands are of orders n and $(n+1)$, the lower order n corresponds to λ_1 and the higher order $(n+1)$ corresponds to λ_2

$$\therefore 2\mu t \cos r = n\lambda_1$$

$$2\mu t \cos r = (n+1)\lambda_2$$

$$n\lambda_1 = (n+1)\lambda_2$$

$$\therefore n = \frac{\lambda_2}{\lambda_2 - \lambda_1} = 60$$

$\therefore$ we get thickness,

$$t = \frac{n\lambda_1}{2\mu \cos r} = \frac{60 \times 6100 \times 10^{-10}}{2 \times 1.33 \times \cos 36.85} = 1.72 \times 10^{-5} \text{ m}$$

Ans: thickness of the film is $1.72 \times 10^{-5} \text{ m}$

Q8. In newton ring experiment the diameter of 4th and 12th dark rings are 0.4cm and 0.7cm. Deduce the diameter of 20th ring.

Given: - $n=4$, $n+p=12$

$$D_4 = 0.4 \text{ cm} \quad D_{12} = 0.7 \text{ cm}$$

Formula: - $D_n^2 = 4Rn\lambda$, $D_{n+p}^2 = 4(n+p)R\lambda$

Solution: - $D_{12}^2 - D_4^2 = 4 \times 8 \times \lambda R$ (1)

SIMILARLY,

$$D_{20}^2 - D_4^2 = 4 \times 16 \times \lambda R \text{ (2)}$$

Divide eqn (1) by eqn(2),

$$\frac{D_{12}^2 - D_4^2}{D_{20}^2 - D_4^2} = \frac{4 \times 8}{4 \times 16} = \frac{1}{2}$$

$$D_{20}^2 = 2D_{12}^2 - D_4^2 = 2 \times 0.7^2 - 0.4^2$$

$$= 0.98 - 0.16 = 0.82$$

$$\text{Diameter of 20}^{\text{th}} \text{ ring is } = \sqrt{0.82} = 0.906 \text{ cm}$$

Ans: Diameter of 20th ring is $= 0.906 \text{ cm}$

Q9. A thin antireflection film ($\mu_f=1.47$) is coated on the surface of a solar cell ($\mu_s=3.6$) to minimize reflection losses. If the average wavelength of the sunlight is taken as $5500\text{\AA}$. What minimum thickness of the film should be deposited on the solar cell surface.

Given: -

$$\mu_f=1.47, \mu_s=3.6, \mu_a=1,$$

$$\lambda=5000\text{\AA}$$

$$i=0 \therefore r=0$$

Formula: - for Destructive interference

$$\text{Path difference}=(2n-1)\lambda/2$$

As the additional P.D of $\lambda/2$ is introduced at each of the points on the denser surface, the condition becomes:

$$2\mu_f t \cos r \pm \lambda/2 + \lambda/2 = (2n\pm 1)\lambda/2$$

$$\therefore 2\mu_f t \cos r = (2n\pm 1)\lambda/2$$

Solution: -

We have , for $r=0$;

$$2\mu_f t = (2n\pm 1)\lambda/2$$

$$t = (2n\pm 1) \frac{\lambda}{4\mu_f}$$

$$\therefore t_{\min} = \frac{\lambda}{4\mu_f} = \frac{5500 \times 10^{-10}}{4 \times 1.47} = 935 \text{\AA}$$

Ans: minimum thickness is $935 \text{\AA}$.

Q10. A wedge shaped air film having an angle of $40''$ is illuminated by mono chromatic light and fringes are observed vertically through a microscope. The distance measured between two consecutive bright fringes is 0.12cm . Calculate the wavelength of light used

Given: - $\theta = 40 \text{ secs} = \frac{40}{3600} \text{ degrees}$

$$= \frac{40 \times \pi}{3600 \times 180} \text{ radians}, \quad \beta = 0.12 \text{ cm}$$

Formula: - $\beta = \frac{\lambda}{2\theta}$

Solution: - spacing between the consecutive bright fringes is,

$$\beta = \frac{\lambda}{2\theta} \quad (\text{for air film})$$

The wavelength is, $\lambda = 2\beta\theta = 2 \times 0.12 \times \frac{40 \times \pi}{3600 \times 180}$

$$= 4654 \times 10^{-8} \text{ cm}$$

$$\lambda = 4654 \text{ \AA}$$

Ans: The wavelength of light used is 4654 \AA

Q12. In costume jewellery, rhinestones (made of the glass with $\mu_r = 1.5$) are often coated with silicon monoxide ($\mu_s = 2$) to make them more reflective. How thick should be the coating to achieve strong reflection for 560 nm light incident normally?

Given: -

$$\mu_r = 1.5, \mu_s = 2, \mu_a = 1,$$

$$\lambda = 560 \text{ nm}$$

$$i = 0 \therefore r = 0$$

Formula: - for strong reflection,

$$p.d = n\lambda$$

Solution: - due to reflection at the denser surface,

$$p.d = 2\mu_s t \cos r \pm \lambda/2$$

$$\begin{aligned}\therefore 2\mu_s t \pm \lambda/2 &= n\lambda \\ \therefore 2\mu_s t &= (2n \pm 1)\lambda/2\end{aligned}$$

$$t = (2n-1) \frac{\lambda}{2\mu_s} = \frac{560 \times 10^{-9}}{4 \times 2} = 70 \text{ nm}$$

Ans: minimum thickness=70nm

Q13. Two optically plane glass strips of length 10cm are placed one over the other. A thin foil of thickness 0.01mm is introduced between them at one end to form an air film. If the light used has wavelength $5900 \text{ \AA}$. Find the width of consecutive bright fringes.

Given: - $l = 10 \text{ cm} = 0.1 \text{ m}$, $t = 0.01 \times 10^{-3} \text{ m}$, $\mu = 1$, $\lambda = 5.9 \times 10^{-7} \text{ m}$

Formula: - $\beta = \frac{\lambda}{2\mu\alpha}$

Solution: -

$$\alpha = \frac{t}{l} = \frac{0.01 \times 10^{-3}}{0.1} = 10^{-4} \text{ radian}$$

$$\beta = \frac{\lambda}{2\mu\alpha} = \frac{5.9 \times 10^{-7}}{2 \times 1 \times 10^{-4}}$$

$$= 2.95 \times 10^{-3} \text{ m}$$

$$\beta = 2.95 \text{ mm.}$$

Ans: The width of consecutive bright fringes is 2.95mm

Q14. In a Newton's rings experiment the diameter of the 15th ring was found to be 0.59 cm and that of the 5th ring is 0.336 cm. If the radius of curvature of the lens is 100 cm, find the wave length of the light.

Given:-

Diameter of Newton's 15th ring (D_{15}) = 0.59 cm = 0.59×10^{-2} m

Diameter of Newton's 5th ring (D_5) = 0.336 cm = 0.336×10^{-2} m

Radius of curvature of lens (R) = 100 cm = 1 m

Formula: - $\frac{D_{n+m}^2 - D_n^2}{4mR}$

Solution: -

$$\begin{aligned} &= \frac{D_{n+m}^2 - D_n^2}{4mR} = \frac{(0.59 \times 10^{-2})^2 - (0.336 \times 10^{-2})^2}{4 \times 1 \times 10} \\ &= \frac{0.235204 \times 10^{-4}}{40} \\ &= 5880 \times 10^{-10} = 5880 \text{Å} \end{aligned}$$

Ans: The wavelength of light is 5880Å

Q15. Two planes glass surfaces in contact along one edge are separated at the opposite edge by a thin wire. If 20 interference fringes are observed between these the sodium light of wavelength of 5893Å at normal incidence what is the thickness of the wire?

Given: -

$$\lambda = 5893 \text{Å} = 5893 \times 10^{-8} \text{ cm}$$

No of fringes seen = 20

$$\mu = 1$$

d = diameter of the wire

Formula: -

$$\beta = \frac{\lambda}{2\mu\alpha}$$

Solution: - For a very small α ,

$$\alpha = \tan \alpha = \frac{d}{x} \text{ and } x = 20\beta$$

$$d = x \alpha = 20\beta \alpha = 20\beta \times \frac{\lambda}{2\mu\beta}$$

$$= \frac{10\lambda}{\mu} = \frac{10 \times 5893 \times 10^{-8}}{1}$$

$$d = 58930 \text{ \AA}$$

Ans: The diameter is $58930 \text{ \AA}$

Q16. Light of wave length 500 nm forms an interference pattern on a screen at a distance of 2 m from the slit. If 100 fringes are formed within a distance of 5 cm on the screen, find the distance between the slits.

Given: -

$$\text{Wave length of light } (\lambda) = 500 \text{ nm} = 500 \times 10^{-9} \text{ m}$$

$$\text{Distance of screen } (D) = 2 \text{ m}$$

Formula: -
$$\beta = \frac{D\lambda}{2d}$$

Solution: -

$$\text{Fringe width } (\beta) = \frac{5}{100} \text{ cm} = \frac{50}{100} \text{ mm} = 0.5 \text{ mm}$$

$$= 0.5 \times 10^{-3} \text{ m}$$

Separation between slits (2d) = ?

$$\beta = \frac{D\lambda}{2d}$$

$$2d = \frac{\lambda D}{\beta} = \frac{500 \times 10^{-9}}{0.5 \times 10^{-3}} = 2 \times 10^{-3} m = 2 mm$$

Ans: The fringe width is $0.5 \times 10^{-3} m$ and separation between slit is $2 mm$

Q17. What will be the order of the dark ring which will have double the diameter of the 40th dark ring?

Given: - $D_n = 2D_{40}$

Solution: -

$$D_{40}^2 = 4R \times 40 \times \lambda = 160R\lambda$$

$$D_n^2 = 2D_{40}$$

$$4Rn\lambda = 4 \times 160\lambda$$

$$n = 160$$

Ans: The order will be 160.